

Area, Perimeter & Volume

Quick reference sheet · GCSE Maths, Geometry & Measures · keep handy for revision

1. Rectangles, triangles & parallelograms

Shape	Area	Perimeter
Rectangle	length $\times$ width	$2 \times (\text{length} + \text{width})$
Triangle	$\frac{1}{2} \times \text{base} \times \text{height}$	sum of the three sides
Parallelogram	base $\times$ height	sum of the four sides

For triangles and parallelograms, always use the **perpendicular** (vertical) height — never a slanted side.

2. Trapeziums

A trapezium has one pair of parallel sides, a and b , separated by perpendicular height h .

$$\text{Area} = \frac{1}{2} \times (a + b) \times h$$

Example: $a = 6 \text{ cm}$, $b = 10 \text{ cm}$, $h = 4 \text{ cm}$ $\rightarrow$ area $= \frac{1}{2} \times 16 \times 4 = 32 \text{ cm}^2$

3. Circles

Quantity	Formula
Diameter	$d = 2 \times r$
Area	πr^2
Circumference	πd or $2\pi r$

Example: $r = 5 \text{ cm}$ $\rightarrow$ area $= \pi \times 5^2 = 78.5 \text{ cm}^2$ (1 d.p.) · circumference $= 2\pi \times 5 = 31.4 \text{ cm}$ (1 d.p.)

Watch out for questions that give the diameter instead of the radius — halve it first before squaring for the area.

4. Volume of prisms & cylinders

Solid	Volume
Cuboid	length $\times$ width $\times$ height
Prism (any cross-section)	cross-sectional area $\times$ length
Cylinder	$\pi r^2 \times \text{height}$

A cylinder is a prism with a circular cross-section, and a cuboid is a prism with a rectangular cross-section — the same rule (cross-sectional area $\times$ length) applies to all of them.

Example (triangular prism): cross-section base 6 cm, height 4 cm, prism length 10 cm $\rightarrow$ area = $\frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2 \rightarrow$ volume = $12 \times 10 = \mathbf{120 \text{ cm}^3}$

Example (cylinder): $r = 3 \text{ cm}$, height = 8 cm $\rightarrow$ volume = $\pi \times 3^2 \times 8 = \mathbf{226.2 \text{ cm}^3}$ (1 d.p.)

5. Common mistakes

Using a slanted side instead of the perpendicular height for triangles, parallelograms and trapeziums.

Forgetting to halve the diameter to get the radius before using it in a circle or cylinder formula.

Confusing area (cm^2 -type units) with volume (cm^3 -type units) — check the units the question expects.

Rounding too early in a multi-step calculation, which makes the final answer inaccurate — only round the final answer.